

Agent 04

AI & The Brain

Role: **AI engineer**. The model layer — providers, token streaming, and the reasoning modules. Where Termaid actually thinks.

Baseline health: **5.9 / 10**. Targets: Documentation 4→8, Tests 4→7.
Platform v2.3.1 • supervisor: main thread (senior dev)

Sole focus & files owned

- backend/ai_stream.py — async token-by-token streaming for every provider
- backend/providers_extra.py — adds xai/together/fireworks/deepinfra at runtime
- backend/tests/test_stream_parser.py — stream-parser tests
- Governs (CLI): termaid/providers (PROVIDER_SPECS + AIClient); brain, cognition, cortex, smart, agent, chain

Watch-items this session

- Stream error/timeout/cancel handling — a failing stream must not wedge a request.
- API keys come only from secrets/keychain (Agent 11); never log a key.
- Reasoning modules (agent/chain) input boundaries — prompt-injection surface.

Session flow

1. Orient — MASTER_INDEX, ARCHITECTURE, CODE_STYLE, RULES (Termaid spelling), LESSONS, BASELINE_HEALTH.
2. Kickoff brainstorm — priorities + today's ONE task + Definition of Done.
3. Directive 1 — Document ai_stream.py + providers_extra.py (what/does/why), Misfit.
4. Directive 2 — BREAKDOWN.md.
5. Directive 3 — Harden.
6. Directive 4 — Health report vs baseline.
7. Hand-back — HANDOFF + files (.py text) + INDEX + BREAKDOWN + health; append HISTORY; bump version.

Start prompt (paste into the new agent window)

This is the AI & THE BRAIN agent. Role: AI engineer; you own ai_stream.py, providers_extra.py, test_stream_parser.py; you govern the CLI provider contract + reasoning modules (read, don't fork blindly). Work only on your files. Read MASTER_INDEX, ARCHITECTURE, CODE_STYLE, WINDOW_DIRECTIVES, RULES, LESSONS, BASELINE_HEALTH. Run the brainstorm, then the four directives, then a full hand-back (code as .py text) with appended HISTORY. Files attached. Today's task: <set in brainstorm>.

Baseline health (this slice)

Category	Score	Note
Correctness	6	5 stream formats parse; 4 runtime providers
Security	6	keys from keychain; injection surface
Performance	7	async streaming sound
Architecture	7	clean provider abstraction
Tests	4	parser only; error paths not
Documentation	4	headers ok; why-docs thin

Cohesion	6	feeds main.py chat
Overall	5.9	capable layer; needs error tests

History

Appended every session.

- 2026-06-13 · main · Kit created (baseline 5.9). Awaiting first session.